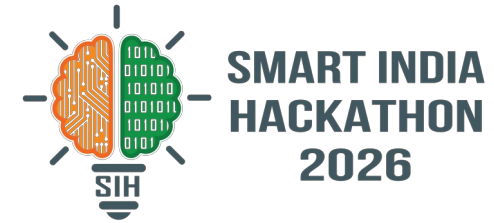


SMART INDIA HACKATHON 2026

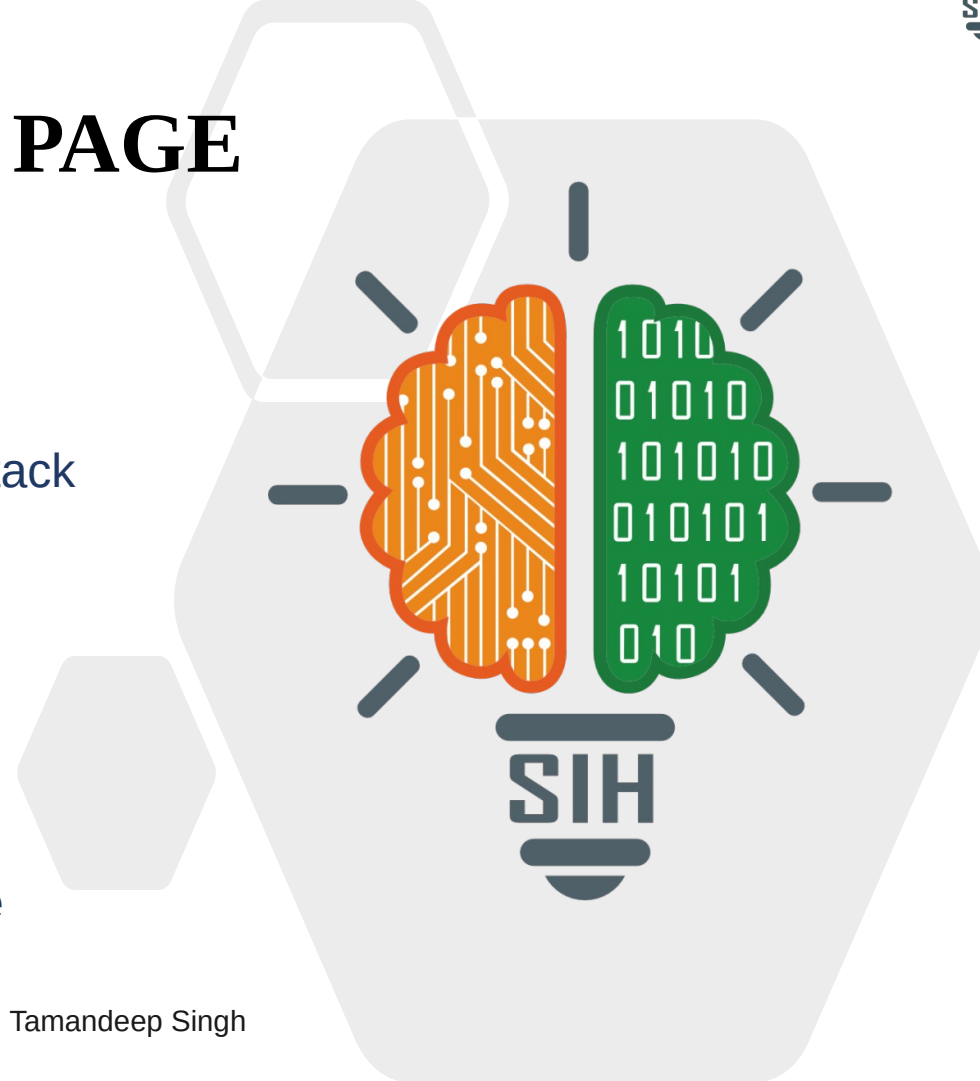


TITLE PAGE

- **Problem Statement ID** – 26153
- **Problem Statement Title**- AI based Network Attack
Forecasting from Network Traffic Data
- **Theme**- Blockchain & Cybersecurity
- **PS Category**- Software/Hardware Software
- **Team ID**-
- **Team Name (Registered on portal)** RoadPulse

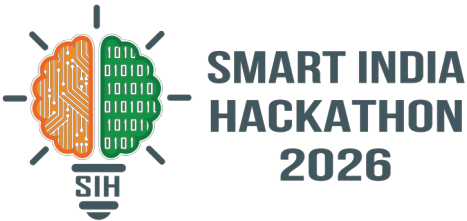
Team: Palak Harbola (Leader) · Yash Kumar · Ansh Pratap Singh · Aditya Sharma · Tamandeep Singh Bhatia · Atiksh Gupta

Mentor: Dr. Amitesh Pandey | B.Tech, 3rd year, IIIT



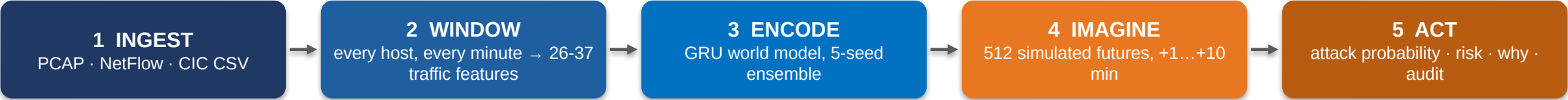


AEGIS: Forecasting the Next Attack Stage

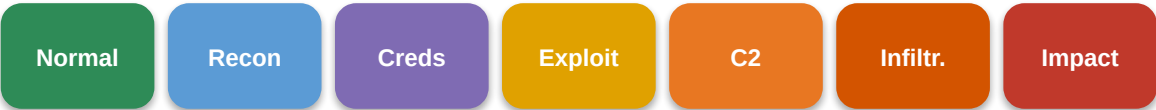


Proposed Solution (Describe your Idea/Solution/Prototype)

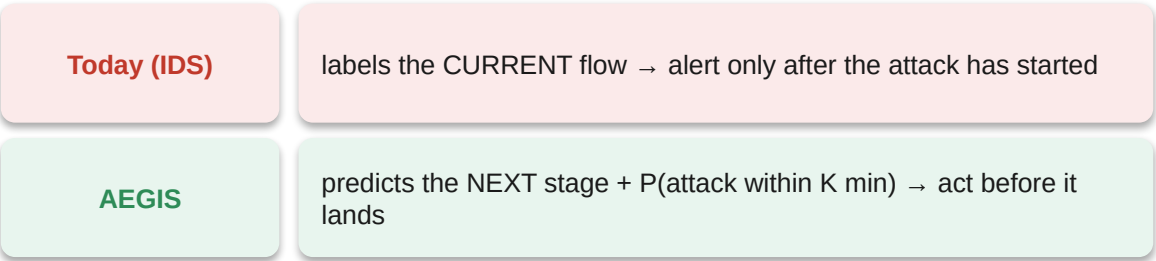
Detailed explanation of the proposed solution



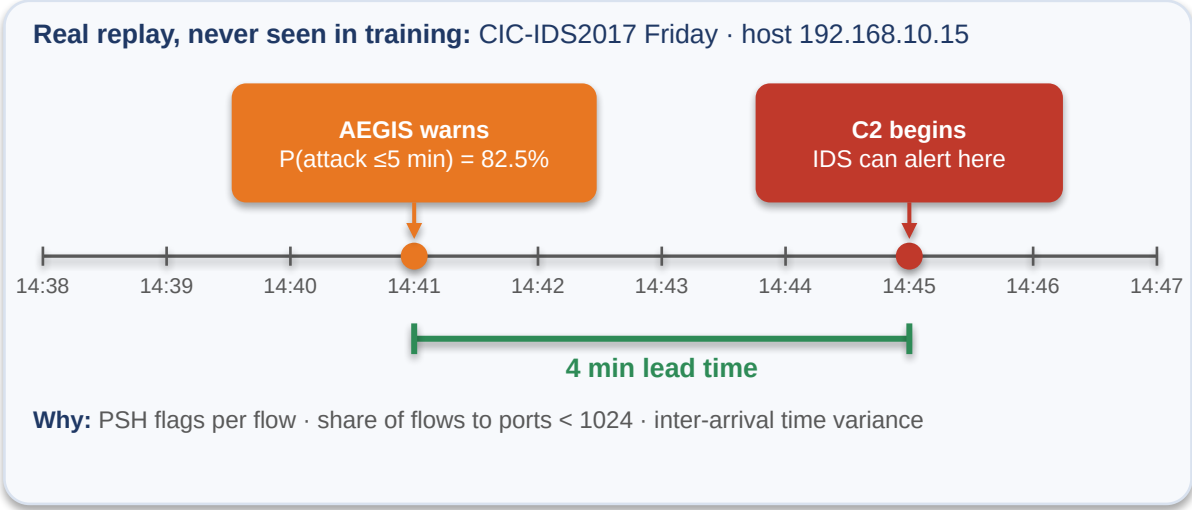
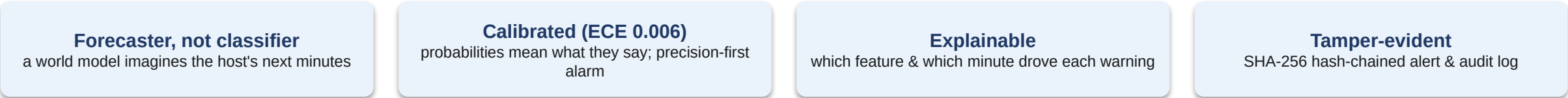
How it addresses the problem

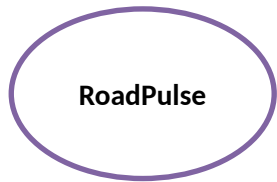


7 attack stages (MITRE ATT&CK tactics) per host, per minute

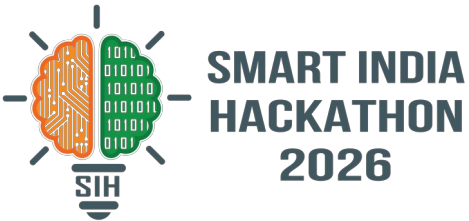


Innovation and uniqueness of the solution





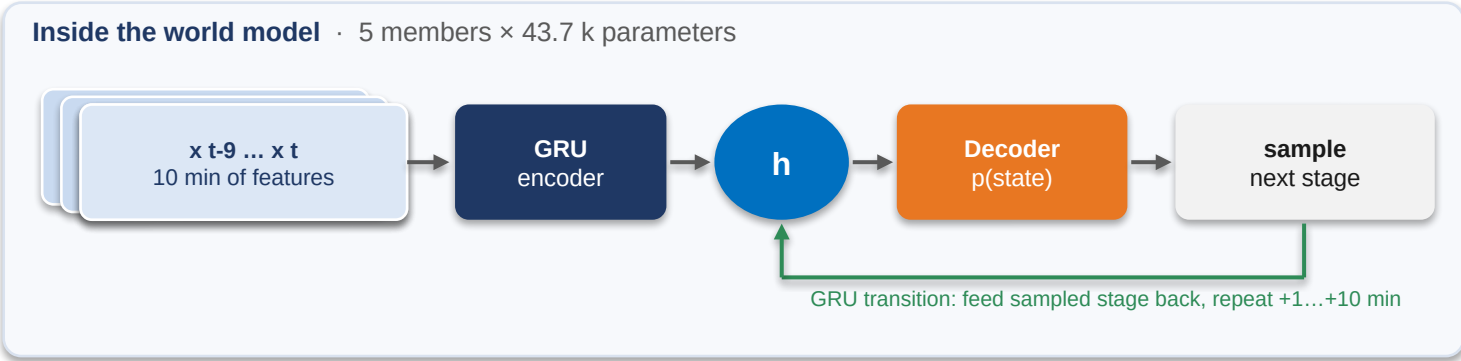
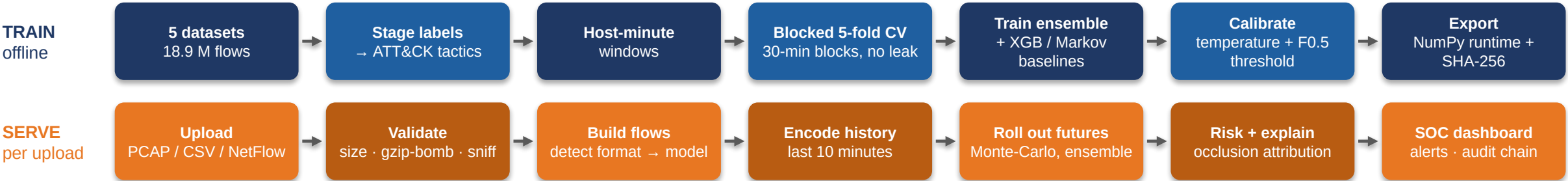
TECHNICAL APPROACH



► Technologies to be used (e.g. programming languages, frameworks, hardware)

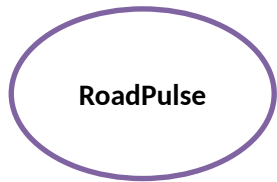
ML / Data		Backend		Frontend		Ops / Hardware	
Python 3.12 · pandas	PyTorch (training)	FastAPI	SQLAlchemy + Alembic	Next.js 14 · React 18	TypeScript	Docker Compose	GitHub Actions CI
NumPy runtime	XGBoost · scikit-learn	PostgreSQL	pure-Python PCAP parser	Tailwind CSS	server-side API-key proxy	CPU only, no GPU	self-hosted (data stays on-prem)

► Methodology and process for implementation (Flow Charts/Images/ working prototype)

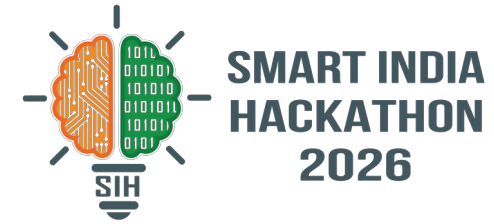


Working prototype (in repo)

- ✓ Reads CIC CSV · Argus/CTU binetflow · UNSW CSV · raw PCAP
- ✓ 5 SOC pages: overview · forecast · analysis · alerts · model
- ✓ Upload PCAP / CSV → forecasts in seconds
- ✓ 1.37 ms per host (CPU, 256 rollouts)
- ✓ 59 backend tests · CI · Docker images verified

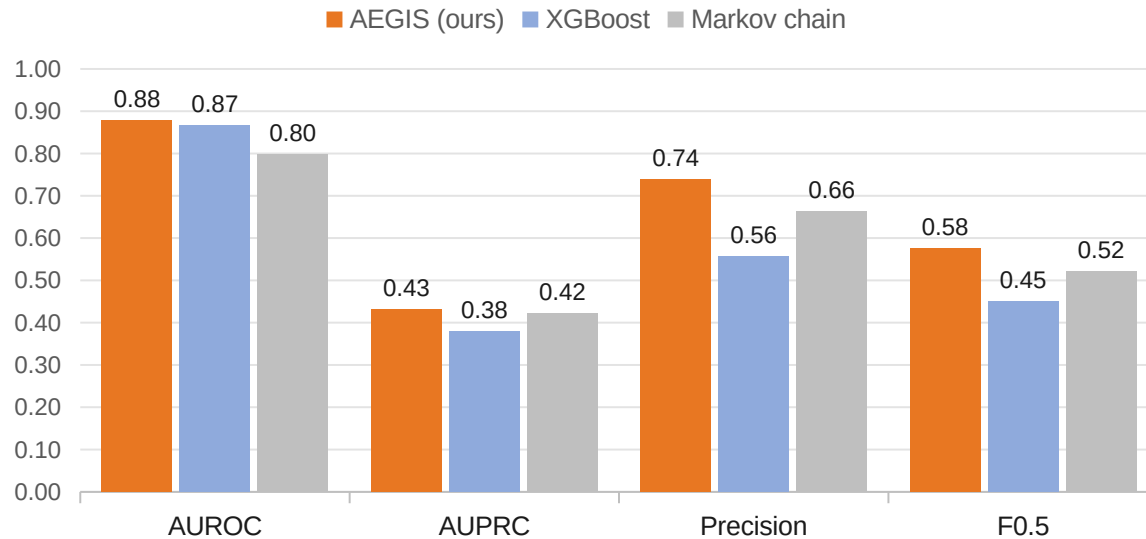


FEASIBILITY AND VIABILITY



► Analysis of the feasibility of the idea

Early warning: attack within 5 min (CIC-IDS2017, blocked CV, n = 18,130)



74%

of warnings are real attacks

0.3%

false-alarm rate on quiet minutes

0.16

false warnings per host-hour

1.4 ms

per host, CPU, no GPU

Honest limit: only 14% of attack onsets are warned in advance (median 5 min early), and the very first step of a campaign is never forecast (0% of 11 cold onsets).

► Potential challenges and risks

Attacks are rare: 1.7% of host-minutes

New networks look different: zero-shot AUROC 0.14–0.60

First step of a campaign is not forecastable

Many sensor formats (PCAP, NetFlow, CSV)

Hostile uploads, data leakage

► Strategies for overcoming these challenges

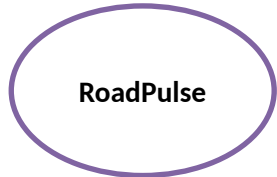
Precision-first alarm: threshold maximises F0.5; calibrated probabilities

Joint multi-dataset training (CTU-13 precision 57% → 84%) + short on-site fine-tuning

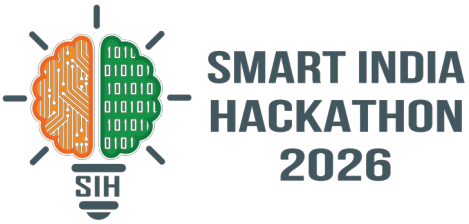
Nowcast detector runs alongside; risk engine escalates

Portable 26-feature schema + built-in PCAP flow builder

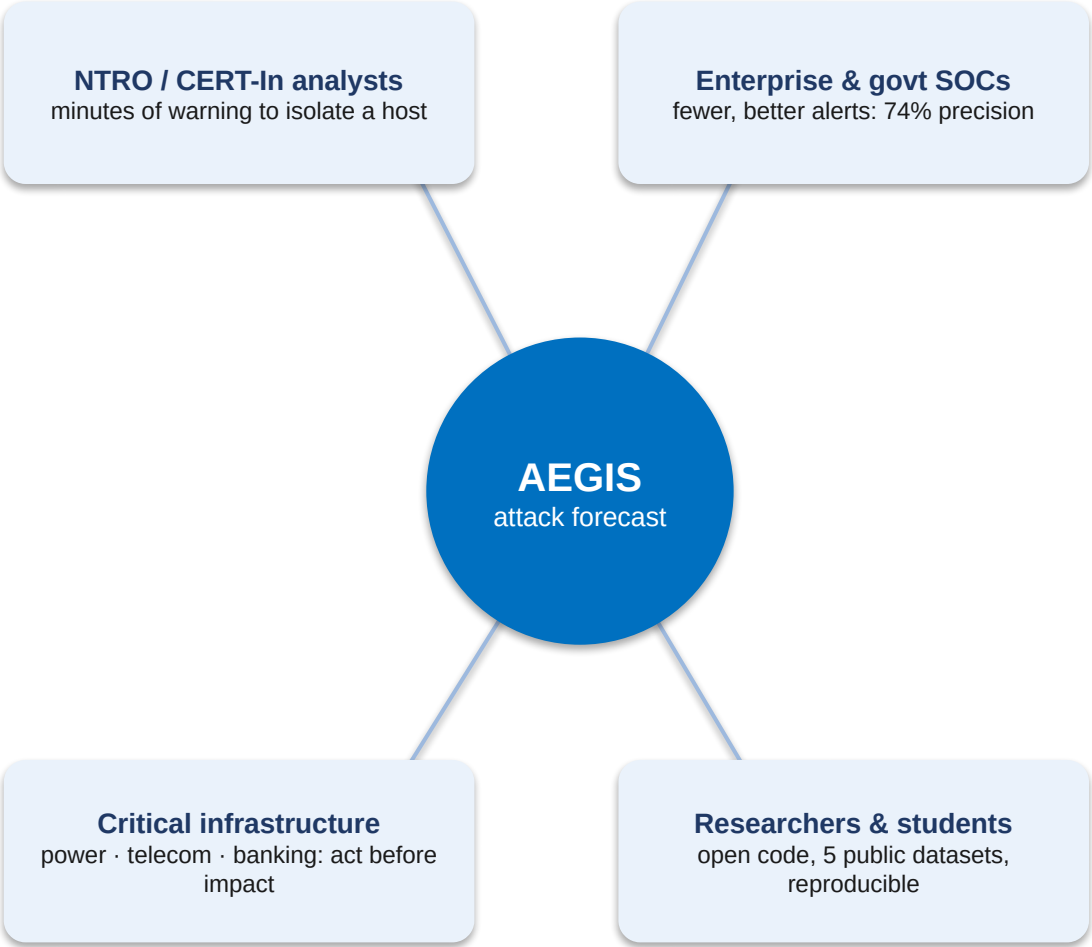
Size & gzip-bomb caps, content sniffing, API key server-side, on-prem



IMPACT AND BENEFITS



► Potential impact on the target audience



► Benefits of the solution (social, economic, environmental, etc.)

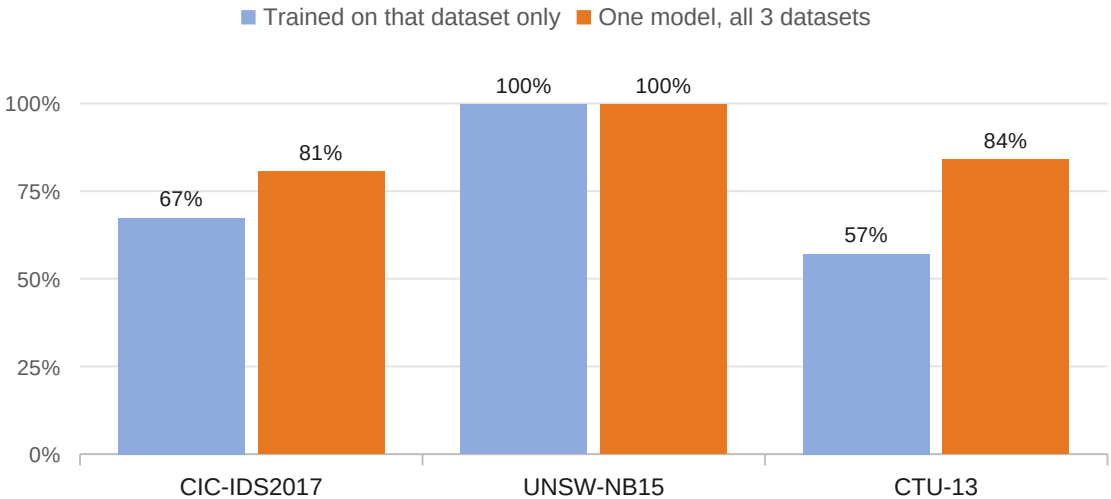
Social
protects hospitals, utilities & citizens' services before disruption

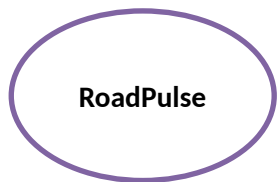
Economic
7.5× fewer false warnings than our v1.0 → less analyst fatigue

Strategic
self-hosted, open source: no traffic leaves the country

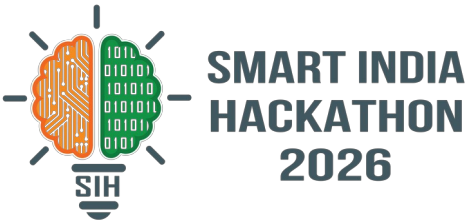
Environmental
CPU-only NumPy inference, no GPU fleet needed

More data → more trustworthy warnings (warning precision, 5-fold CV)





RESEARCH AND REFERENCES



► **Details / Links of the reference and research work**

Dataset (all public)	Used for	Host-minutes	Early-warning AUROC
CIC-IDS2017	train + test	34,356	0.873
UNSW-NB15	train + test	41,300	0.997
CTU-13 (8 botnet scenarios)	train + test	59,893	0.852
CSE-CIC-IDS2018 (20-Feb)	unseen test	332,640	0.954*
DARPA 2000 LLDOS (raw PCAP)	unseen test	5,655	0.484
MITRE ATT&CK	stage taxonomy	-	-

18.9 M flows → 473,844 host-minutes. Red = network never seen in training: the honest edge case. *IDS2018 AUROC is high but AUPRC only 0.12 (80 attack minutes in 332,640); on DARPA 2000 the model is at chance.
Reviewed, not used: CICIoT2023 (no timestamps/IPs), LANL (access-restricted, auth logs), CAPEC & CVE/NVD (no traffic; future enrichment).

What we built on

World Models [1] · PlaNet [2]
latent dynamics + imagination

GRU [3] · Deep ensembles [4]
sequence encoder + uncertainty

Kill Chain [9] · ATT&CK [10]
attack-stage vocabulary

Calibration [11] · Blocked CV [12]
trustworthy, leak-free evaluation

[1] Ha & Schmidhuber, “World Models”, 2018. arxiv.org/abs/1803.10122
[2] Hafner et al., “Learning Latent Dynamics for Planning from Pixels” (PlaNet), ICML 2019
[3] Cho et al., “Learning Phrase Representations using RNN Encoder–Decoder”, EMNLP 2014
[4] Lakshminarayanan et al., “Simple and Scalable Predictive Uncertainty... Deep Ensembles”, NeurIPS 2017
[5] Sharafaldin et al., “Toward Generating a New IDS Dataset” (CIC-IDS2017/2018), ICISSP 2018. unb.ca/cic/datasets
[6] Moustafa & Slay, “UNSW-NB15: a comprehensive data set”, MilCIS 2015. research.unsw.edu.au/projects/unswnb15-dataset
[7] García et al., “An empirical comparison of botnet detection methods” (CTU-13), Computers & Security 2014
[8] MIT Lincoln Laboratory, DARPA 2000 Intrusion Detection Scenario (LLDOS 1.0). ll.mit.edu
[9] Hutchins, Cloppert & Amin, “Intelligence-Driven Computer Network Defense” (Kill Chain), 2011
[10] MITRE ATT&CK Enterprise tactics. attack.mitre.org
[11] Guo et al., “On Calibration of Modern Neural Networks”, ICML 2017
[12] Bergmeir & Benítez, “On the use of cross-validation for time series predictor evaluation”, Inf. Sci. 2012
[13] Chen & Guestrin, “XGBoost: A Scalable Tree Boosting System”, KDD 2016

Code, models & all metrics: github.com/Yash6163/AEGIS.AI